

Fine-Tuned Open-Weight Language Models for Explainable Energy-Theft Detection: Accuracy, Generalization, and Deployment Cost

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Abstract—[EDIT: Draft contains the Experimental Setup and Results sections only; abstract, introduction, related work, and conclusion are written separately.]

Index Terms—Energy theft, non-technical losses, large language models, fine-tuning, smart grid, explainability.

I. EXPERIMENTAL SETUP

Our conference study [1] established that a zero-shot proprietary model can outperform supervised classical baselines on fine-grained energy-theft identification, reaching 0.929 macro-F1 against CatBoost’s 0.881. However, a proprietary API is an impractical operating point for a utility provider. Every query-day is billed at API rates, consumption data leaves the utility’s infrastructure, and the detector cannot be adapted when new manipulation patterns emerge. This section asks whether the zero-shot frontier can be matched by small open-weight models fine-tuned locally. It then asks what such a detector costs, tolerates, and explains. The experiments form a funnel rather than a flat grid; each stage is motivated by the outcome of the one before it. Concretely, we answer six research questions:

- 1) **Parity (RQ1):** Can fine-tuned open-weight models of at most 14B parameters match the zero-shot proprietary frontier and supervised classical baselines on 13-class theft identification?
- 2) **Label and compute cost (RQ2):** If parity holds, how many labeled days and how much training compute does it require, and where does performance saturate?
- 3) **Generalization (RQ3):** If the label bill is affordable, does the detector survive attack types that were absent from its training data?
- 4) **Severity (RQ4):** Where does any detector fail, as a function of the fraction of energy actually removed?
- 5) **Explanation (RQ5):** Beyond detection, can the model state the manipulation mechanism and quantify its magnitude, and at what cost to classification?
- 6) **Deployment (RQ6):** Given the answers above, what does the cost–performance frontier look like against API zero-shot and classical ML, and where is the break-even?

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This publication is supported in part by grant NPRP12C-33905-SP-66 from the Qatar National Research Fund. The findings achieved herein are solely the responsibility of the authors.

The remainder of this section specifies the shared apparatus: the dataset and its frozen partition, the attack taxonomy, the models and their single shared recipe, the baselines, the metrics, and the statistical protocol. The design logic that connects the experiments is stated last, in Section I-F.

A. Task, Dataset, and Data Partitioning

The task is per-day classification of smart-meter consumption. Given one query day of readings together with the consumer’s preceding reference week, the model emits a single JSON object. The object declares the day *original* or *manipulated* and, when manipulated, identifies which of twelve attack mechanisms produced it. The model input is text-serialized numeric data. Each sample contains a metadata line (consumer, month, week, and day identifiers), per-day summary statistics (minimum, maximum, mean, and standard deviation), and calendar features (month, day-of-week, day-of-month). It then carries the raw readings themselves: 96 fifteen-minute values for each of the seven reference days and the query day, 768 numbers in total. No feature engineering beyond this serialization is applied; the model sees what a meter data-management system would store.

All experiments draw from a single dataset of 8,239 labeled consumer-days spanning 324 consumers and all 13 classes. We partition it once, before any experiment, into a consumer-disjoint 80/20 split (seed 42). The 259 training consumers contribute 6,589 days; the remaining 65 consumers contribute the 1,650 evaluation days. Consumer disjointness matters because per-consumer consumption habits are strong identifiers. A random row-level split would leak consumer identity from train to test and inflate every downstream number. The evaluation slice contains 130 benign days and 1,520 manipulated days (105–130 per attack class), an 11.7:1 imbalance. The split is frozen by a manifest recording the dataset’s SHA-256 checksum and every row assignment. The manifest is verified programmatically before each run, so no experiment in this paper saw a different partition.

Supervision targets are complete JSON objects constructed deterministically from the ground-truth labels. Training loss is computed on the completion only, so no capacity is spent reproducing the $\sim 7,900$ -token numeric prompt. At inference the model generates greedily and its output is parsed back to the same schema.

TABLE I
THE 13-CLASS TAXONOMY: THE BENIGN PROFILE PLUS TWELVE ATTACK MECHANISMS, WITH THE FAMILY GROUPING USED FOR THE OUT-OF-CLASS ROTATIONS OF SECTION II-C.

Class	Mechanism	Effect on profile	Family
original	unmodified consumption	—	—
attack0	all samples set to zero for the whole day	removes trend and magnitude	Z
attack1	all samples $\times$ one random factor $\in (0,1)$	trend kept, magnitude reduced	M
attack2	a random contiguous window zeroed (on-off)	changes trend and amplitude	Z
attack3	each sample $\times$ its own random factor $\in (0,1)$	changes trend and amplitude	M
attack4	readings reduced inside a random sub-period	changes trend and amplitude	M
attack5	each sample $\leftarrow$ mean consumption $\times$ random $\in (0,1)$	replaces the pattern	M
attack6	values above a random cut-off clipped to it	changes trend and amplitude	T
attack7	a random cut-off subtracted; negatives to zero	trend kept, total reduced	T
attack9	every sample $\leftarrow$ the day's average (flat profile)	removes all trend	S
attack10	the daily pattern reversed in time	trend changed, total unchanged	S
attack12	profile of a lower-consuming user substituted	false data injection	S
attack13	each interval reports the true value or zero	irregular; trend and amplitude	Z

Families: Z = zero-suppression, M = multiplicative/randomized, T = threshold, S = shape substitution.
attack13 is the IEEE sampled-zeroing attack; attacks 8 and 11 of the source taxonomy are excluded by the data contract.

TABLE II
THE FROZEN FINE-TUNING RECIPE SHARED BY EVERY RUN IN THIS PAPER. ONLY THE AXIS UNDER STUDY VARIES PER EXPERIMENT: THE BASE MODEL (RQ1), THE TRAINING-SET SIZE (RQ2), THE HELD-OUT ATTACK FAMILY (RQ3), THE TARGET SCHEMA (RQ5), AND THE LORA RANK (APPENDIX A).

Knob	Value
Adapter	LoRA (DoRA), $r = 16$, $\alpha = 32$, dropout 0.05
Target modules	all linear layers
Quantization	bf16 (31B model: 4-bit NF4 QLoRA)
Epochs / batch	1 epoch, batch 1×8 gradient-accumulation steps
Learning rate	2×10^{-4} , cosine schedule, 3% warmup
Weight decay	0.01
Max sequence length	8,192–11,008 tokens (tokenizer-dependent)
Decoding at evaluation	greedy, 512 new tokens
Hardware	one NVIDIA H100 80GB

B. Attack Taxonomy

Table I lists the 13-class taxonomy: the benign class plus twelve synthetic attack mechanisms drawn from the non-technical-loss literature [2], [3]. Each mechanism is a distinct transformation of the daily load profile. The taxonomy spans the practically relevant spectrum, from total meter bypass (attack0) to manipulations that leave total energy unchanged and alter only the profile’s shape (attack10). For the generalization experiments of Section II-C we group the twelve mechanisms into four families by manipulation type: zero-suppression, multiplicative or randomized reduction, threshold clipping or subtraction, and shape substitution.

C. Models and Fine-Tuning Configuration

We screen seven open-weight instruction-tuned models spanning 4B to 31B parameters and four model families: Qwen3-8B and Qwen3-14B [4], Phi-4-reasoning (14B) [5], Ministral-3-14B [6], Llama-3.1-8B [7], Gemma-4-E4B (4B effective), and Gemma-4-31B [8]. Every model is fine-tuned with the identical recipe of Table II — LoRA [9] in its weight-decomposed variant [10]; only the base model varies. This single-recipe design is a deliberate trade-off. It makes the screening comparison clean at the risk of disadvantaging models whose optimal recipe differs, and we validate the choice retrospectively in Appendix A. The 31B model exceeds the memory budget for 16-bit adaptation on a single GPU and is trained with 4-bit quantized adaptation [11] instead; we flag this asymmetry where it matters.

Two evaluation-side details are worth stating because they shape what the metrics can see. First, adapters are merged

into the base weights before inference, so evaluation throughput reflects a deployable artifact rather than adapter-wrapped research code. Second, a generative classifier emits a label, not a score, which would leave threshold-based metrics such as AUC undefined. We therefore recover a continuous score by verbalizer scoring: the output is teacher-forced up to the classification field and the first-token logits of the two class words are renormalized. This restores AUC and enables threshold analysis without altering the generated predictions. Finally, a machine-readable hardware manifest exists only for the legacy 32B pilot run. For the journal-round runs the hardware statement above derives from configuration records rather than per-run capture, a provenance gap we note for completeness.

D. Baselines and Reference Bands

Three kinds of comparison appear throughout, and we are careful not to conflate them. The *zero-shot proprietary frontier* is Gemini 3.1 Pro (0.976 binary balanced accuracy, 0.929 macro-F1), reported on the same consumer-disjoint split as every result in this paper; the parity question of RQ1 is therefore a like-for-like comparison. *Refit baselines* are classical models trained and evaluated on exactly our split: a random forest, CatBoost, XGBoost, a logistic regression, and an unsupervised local outlier factor, all consuming the same per-day readings the LLMs see. These support paired comparisons. The tree and boosting baselines use seeded library-default hyperparameters; the conference’s tuned CatBoost configuration is not recorded in our artifacts, and we state this where it matters. *Reference bands* are frozen classical numbers from the conference study — CatBoost, XGBoost, random forest, and LOF — obtained on its different, 1,261-row benchmark. Reference-band rows are marked with a dagger (†) in every table, are never bolded, and are never subjected to statistical tests. They contextualize our numbers but cannot be paired with them. This distinction is stated once here and applies to every table that follows. Of the refit baselines, the three tree-family models carry full metrics; the logistic regression and LOF are recorded as balanced accuracy only.

E. Evaluation Metrics and Statistical Protocol

Classification metrics : Binary detection is reported as balanced accuracy, F1 on the manipulated class, AUC-ROC (via verbalizer scores), specificity, precision, and recall. Identification is reported as macro-F1, weighted-F1, balanced accuracy, and per-class F1 over the 13 classes. We emphasize specificity alongside recall because the two error types carry asymmetric operational cost. A false positive is a site inspection dispatched to an honest consumer, and at utility scale the inspection budget, not the detection rate, is usually the binding constraint.

Statistical protocol : All headline metrics carry 95% percentile confidence intervals from a consumer-level cluster bootstrap over the 65 evaluation consumers ($B = 10,000$, seed 42). Resampling consumers rather than days respects the within-consumer correlation of consumption behavior. Paired model comparisons use the same bootstrap on the metric difference plus McNemar’s test on the per-day predictions,

exact binomial when discordant pairs are fewer than 25, with Holm correction over the seven screening comparisons. Reference-band rows are excluded from all tests, as stated above.

Explanation metrics : Rationale quality is measured by ROUGE-L [12] against the teacher rationale and, more stringently, by parameter-extraction fidelity. For the three attacks whose generation parameters are recoverable from the data (attacks 1, 2, and 4; $n = 375$, about 39% of manipulated evaluation days), we check whether the stated attack parameter is numerically correct. Magnitude estimates (Section II-F) are scored by mean absolute error against the realized energy-reduction fraction, rank correlation, bucket accuracy, and win-dow intersection-over-union.

Severity proxy : Several analyses stratify by theft severity using $\rho = 1 - E_{\text{query}}/\overline{E}_{\text{reference week}}$, the fraction of expected energy removed, with frozen bucket edges at 10%, 30%, and 60%. The benign day-to-day variation of ρ spans -0.197 to $+0.178$ (5th to 95th percentile), so the 30% edge sits outside natural variation. Attacks 9 and 10 do not change the day’s total energy; they only reshape the consumption profile (median $\rho \approx 0$). We call these the *energy-neutral* pair and report them separately rather than forcing them into severity buckets.

F. Experimental Design and Research Questions

The experiments form a funnel. A single screening pass (RQ1) fine-tunes all seven models once under the shared recipe and applies a selection rule that was registered before the screening ran. The *flagship* is the model with the best macro-F1. The *workhorse* is the smallest model whose paired-bootstrap 95% confidence interval on the macro-F1 gap to the flagship contains zero, with lower inference cost as the tie-break. Every subsequent experiment — the data-efficiency curve (RQ2), the unseen-attack rotations (RQ3), the rationale distillation and magnitude extensions (RQ5), and the rank validation — runs on the workhorse only. The severity analysis (RQ4) and the deployment frontier (RQ6) require no new training; they are computed from the predictions of the runs already made. Pre-registering the rule matters because it removes the selection of the downstream model from our own discretion. The screening result licenses the cut, not our preference. **[EDIT: Funnel diagram (S2 → selection rule → S3/S4/S6/F018 on the workhorse; S5/S8 prediction-level) — TikZ or draw.io; the draft ships without it.]**

II. RESULTS AND DISCUSSION

In this section, we present the results of the experiments and discuss them in the order the funnel dictates. First, the screening pass answers whether fine-tuned open models reach the zero-shot frontier and selects the two models that carry the rest of the paper. Then, we cost the result in labels and compute, and stress it against attack families the model never saw. Next, the failure mechanism exposed by that stress test is formalized into a severity analysis. The severity finding sharpens the question of what an LLM offers that

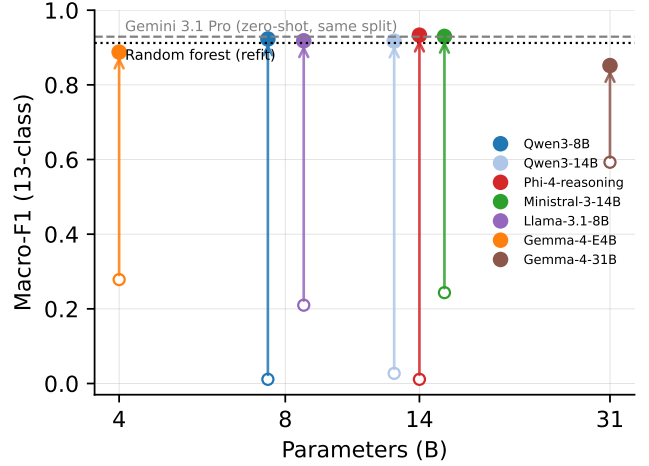


Fig. 1. Macro-F1 against parameter count before and after fine-tuning. Each arrow joins a model’s zero-shot score to its fine-tuned score; same-size models are dodged horizontally for legibility. The dashed band is the zero-shot Gemini 3.1 Pro frontier (same split) and the dotted band the refit random forest.

a much cheaper classical model does not, and the rationale-distillation and magnitude experiments answer it. Lastly, the deployment frontier assembles accuracy, cost, and capability into an operating recommendation.

A. Screening and Model Selection (RQ1)

This experiment establishes the premise of the paper: whether fine-tuned open-weight models at practical scales reach the zero-shot proprietary frontier that motivated this work. Table III reports all seven models in both their zero-shot and fine-tuned conditions on the 1,650-day evaluation slice, against the refit baselines and the frozen conference bands. Figure 1 visualizes the fine-tuning lift per model.

The answer to RQ1 is affirmative. The best fine-tuned model, Phi-4-reasoning, reaches 0.9331 macro-F1, which exceeds Gemini 3.1 Pro’s 0.929 on the same split, and four of the seven models land within 0.016 of it. However, the zero-shot rows show that none of this competence is present before fine-tuning, and the failures are instructive because they are not uniform. Qwen3-8B and Phi-4-reasoning predict *original* for essentially every day (F1 of 0.0000 at a specificity of 1.0). Llama-3.1-8B and Gemma-4-E4B collapse in the opposite direction and declare nearly every day manipulated (specificity of 0.0231 and 0.0154). Notably, zero-shot Qwen3-8B attains an AUC of 0.9066 while its F1 is exactly zero. The ranking signal is already present in the base model; what is missing is only the decision boundary. This suggests that fine-tuning is not teaching the model to see theft so much as teaching it where to draw the line, an interpretation the data-efficiency curve of Section II-B will support directly. The single genuine zero-shot competitor is Gemma-4-31B at 0.5927 macro-F1. Yet the same model, fine-tuned, is the *worst* of the seven at 0.8517, receiving the smallest lift from adaptation — though this model is also the only one adapted in 4-bit, so its poor showing conflates scale with quantization and we do not read a scale conclusion from it. The converse case carries

TABLE III

MAIN RESULTS ON THE 1,650-DAY CONSUMER-DISJOINT EVALUATION SLICE, IN FIVE BLOCKS. THE GEMINI FRONTIER IS REPORTED ON THE SAME SPLIT; ROWS MARKED † ARE FROZEN CONFERENCE-BENCHMARK CLASSICAL REFERENCES (1,261-ROW BENCHMARK): CONTEXT BANDS, NOT PAIRED COMPARISONS. BEST VALUE PER COLUMN IS BOLDDED WITHIN THE FINE-TUNED BLOCK ONLY.

Model		Binary detection				13-class identification		
		Bal-acc	F1	AUC	Specificity	Macro-F1	Weighted-F1	Bal-acc
Zero-shot proprietary	Gemini 3.1 Pro (same split)	0.976	—	—	—	0.929	—	—
Zero-shot open	Qwen3-8B	0.5000	0.0000	0.9066	1.0000	0.0112	—	0.0769
	Qwen3-14B	0.5053	0.0208	0.9193	1.0000	0.0271	—	0.0858
	Phi-4-reasoning	0.5000	0.0000	0.5863	1.0000	0.0112	—	0.0769
	Ministral-3-14B	0.6893	0.5648	0.9473	0.9846	0.2434	—	0.2986
	Llama-3.1-8B	0.5059	0.9543	0.8425	0.0231	0.2098	—	0.2885
	Gemma-4-E4B	0.5041	0.9560	0.8747	0.0154	0.2784	—	0.3842
	Gemma-4-31B	0.9137	0.9767	0.9841	0.8615	0.5927	—	0.6515
Fine-tuned	Qwen3-8B	0.9406	0.9898	0.9939	0.8923	0.9229	0.9247	0.9210
	Qwen3-14B	0.9541	0.9928	0.9947	0.9154	0.9173	0.9188	0.9154
	Phi-4-reasoning	0.9502	0.9924	0.9964	0.9077	0.9331	0.9341	0.9321
	Ministral-3-14B	0.9569	0.9921	0.9972	0.9231	0.9298	0.9307	0.9287
	Llama-3.1-8B	0.9191	0.9895	0.9939	0.8462	0.9181	0.9197	0.9177
	Gemma-4-E4B	0.9456	0.9878	0.9941	0.9077	0.8878	0.8893	0.8850
	Gemma-4-31B	0.9556	0.9800	0.9890	0.9462	0.8517	0.8537	0.8503
Supervised ML	Random forest (refit)	0.8475	—	—	—	0.9122	—	—
	CatBoost (refit)	0.8145	0.9765	0.9828	0.6462	0.9093	—	0.9098
	XGBoost (refit)	0.8202	0.9751	0.9811	0.6615	0.9076	—	0.9075
	Logistic regression (refit)	0.8669	—	—	—	—	—	—
	CatBoost†	0.8599	0.9823	0.9855	0.7330	0.8806	—	0.8809
	XGBoost†	0.8698	0.9820	0.9857	0.7554	—	—	—
Unsupervised	LOF (refit)	0.8209	—	—	—	—	—	—
	LOF†	0.5490	0.2817	0.6278	0.9332	—	—	—

Refit baselines are trained on the identical 6,589/1,650 split; of these only the random forest has a recorded macro-F1.

TABLE IV

THE PRE-REGISTERED SELECTION RULE APPLIED TO THE SCREENING RESULTS: PAIRED CONSUMER-CLUSTER BOOTSTRAP (65 CLUSTERS, $B = 10,000$) ON THE MACRO-F1 GAP TO THE FLAGSHIP. THE WORKHORSE ROW IS BOLDDED.

Model	Params (B)	Macro-F1	Δ vs. flagship	95% CI	Tied
Phi-4-reasoning	14	0.9331	—	—	flagship
Ministral-3-14B	14	0.9298	−0.0033	[−0.0141, +0.0072]	yes
Qwen3-8B	8	0.9229	−0.0102	[−0.0225, +0.0021]	yes
Llama-3.1-8B	8	0.9181	−0.0149	[−0.0284, −0.0022]	no
Qwen3-14B	14	0.9173	−0.0158	[−0.0286, −0.0036]	no
Gemma-4-E4B	4	0.8878	−0.0453	[−0.0576, −0.0337]	no
Gemma-4-31B	31	0.8517	−0.0814	[−0.1000, −0.0631]	no

no such confound: Phi-4-reasoning enters with the weakest zero-shot signal of all (a verbalizer AUC of 0.5863, barely above chance) and leaves as the flagship. In other words, zero-shot competence and fine-tuned competence are different properties, and neither predicts the other in either direction.

Table IV applies the pre-registered rule. Two models are statistically tied with the flagship, and the smaller of them, Qwen3-8B, becomes the workhorse. It trails the flagship by 0.0102 macro-F1 with a 95% confidence interval of [−0.0225, +0.0021] that contains zero. One anomaly deserves a sentence: Qwen3-14B trails its own 8B sibling by 0.0056 macro-F1 and is *not* tied with the flagship. Within this family, scale did not buy identification quality under the shared recipe. Per-class results (Figure 2; full table in Appendix B) show the same two classes limiting every model. These are attack1, the

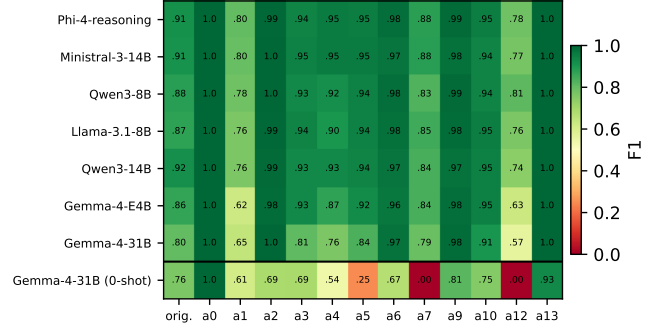


Fig. 2. Per-class F1 for the seven fine-tuned models, with the best zero-shot model (Gemma-4-31B) as the bottom row for contrast. attack1 and attack12 limit every model.

uniform multiplicative reduction that most resembles a frugal consumer, and attack12, the consumption-profile substitution, with per-class F1 as low as 0.78 and 0.81 for the workhorse against a near-perfect 1.00 on attack0 and attack13. A rank sweep around the shared recipe (Appendix A) finds $r = 8$ statistically tied with the default and $r = 32$ significantly worse, so the screening was not capacity-limited and the single-recipe design was fair.

One number in Table III, however, disciplines every claim that follows. The refit random forest reaches 0.9122 macro-F1

TABLE V

IDENTIFICATION AND DETECTION QUALITY AGAINST TRAINING-SET SIZE FOR THE WORKHORSE, WITH THE MATCHED RANDOM-FOREST REFITS (CLASS-STRATIFIED SUBSETS; SEED MEANS OVER THREE DATA SEEDS AT $N=65$ AND $N=260$). COST ASSUMES AN H100 AT \$2.50/H; THE FOREST TRAINS IN CPU-MINUTES AT EVERY SIZE.

N (per class)	Macro-F1	RF macro-F1	Bin. bal-acc	AUC	Specificity	Train time	Cost
65 (5)	0.2588	0.7451	0.4787	0.8869	0.008	3.1 min	\$0.13
260 (20)	0.3540	0.8321	0.5061	0.9112	0.049	12.5 min	\$0.52
1,040 (80)	0.7488	0.8855	0.7838	0.9711	0.585	49.7 min	\$2.07
6,589 (full)	0.9229	0.9122	0.9406	0.9939	0.892	5.23 h	\$13.08

Seed-level macro-F1 — workhorse, $N=65$: 0.2427/0.2687/0.2650; $N=260$: 0.3620/0.3690/0.3310. RF, $N=65$: 0.7905/0.7327/0.7120; $N=260$: 0.8473/0.8324/0.8167. CatBoost seed means: 0.7667/0.8624/0.8915; XGBoost: 0.6919/0.8380/0.8790 (per N , ascending). Bold marks the better identifier per row. AUC and specificity at $N \leq 260$ are workhorse seed means.

on the identical split, from the same readings, at effectively zero inference cost; CatBoost and XGBoost refits land just below it (0.9093 and 0.9076), so the forest is the classical ceiling here and the tests that follow are against the strongest classical baseline. Paired on the same days, the workhorse’s identification advantage over it is not statistically significant ($\Delta = +0.0107$, 95% CI $[-0.0055, +0.0265]$; McNemar $p = 0.204$); only the flagship separates ($\Delta = +0.0209$, CI $[+0.0051, +0.0368]$, $p = 0.0095$), and it does so against all three classical refits ($p \leq 0.0037$ for CatBoost and XGBoost). What the workhorse does hold over the forest is binary detection: its specificity of 0.892 against the forest’s 0.715 more than halves the false-alarm rate (McNemar $p < 10^{-4}$), which is fewer wasted site inspections, not a better attack taxonomy. If accuracy were the whole story, fine-tuning an LLM would be difficult to justify, and the remainder of this section should be read as an answer to that objection. In summary, we can conclude that fine-tuned open models at 8–14B parameters match the zero-shot proprietary frontier. The pre-registered rule hands every subsequent experiment to the workhorse, Qwen3-8B — beginning with the question of what the result costs in labels and compute.

B. Label and Compute Efficiency (RQ2)

Parity with the frontier is only useful if the labeling cost is payable, since annotated theft days are precisely what utilities lack. This experiment therefore prices the workhorse’s result. We retrain it on stratified subsets of 5, 20, and 80 labeled days per class (three data seeds at the two smallest sizes) and compare against the full 6,589-day run. Table V and Figure 3 report the curve.

The conclusion is read off the curve as follows. Each row of the data-curve table retrains the workhorse from the same base checkpoint, varying only the number of stratified training days; everything else — recipe, evaluation slice, decoding — is held fixed, so any change in macro-F1 is attributable to the labels alone. Growing the set from 65 to 260 days buys 0.095 macro-F1 (about 0.5 points per 1,000 additional labels at this scale); growing it from 260 to 1,040 buys 0.395 (again about 0.5 per 1,000, sustained over a far larger step); growing it from 1,040 to 6,589 buys only 0.174 (0.03 per 1,000). The marginal value of a label collapses after the third point, which is why we place the elbow between 20 and 80 labels per class: at 1,040 days the model already holds 0.7488, or 81% of the full-data 0.9229. The full run costs 5.23 GPU-hours, roughly \$13 at assumed rates, and the 1,040-day point costs \$2.07. Label acquisition, not compute, is clearly the binding budget.

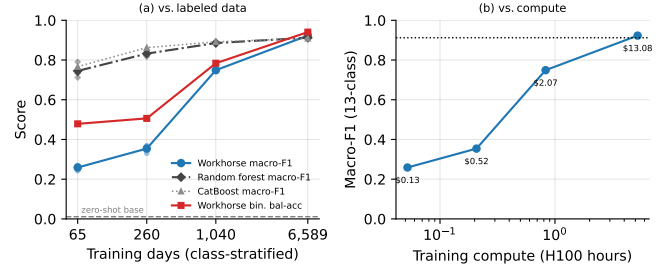


Fig. 3. The data-efficiency curves of the workhorse and the matched classical refits: (a) macro-F1 against training-set size, with the workhorse’s binary balanced accuracy; (b) workhorse macro-F1 against training compute (the classical models train in CPU-minutes at every size). Per-seed points shown at $N=65$ and $N=260$; XGBoost (slightly below the forest throughout) is omitted for legibility and tabulated in Appendix D. Every classical variant dominates until full data.

The matched random-forest refits answer the comparison this curve owes the reader, and the answer is one-sided below full data. Trained on the identical subsets, the forest holds 0.7451 macro-F1 at 65 days, 0.8321 at 260, and 0.8855 at 1,040 — against the workhorse’s 0.2588, 0.3540, and 0.7488. It also reaches a usable decision boundary immediately (specificity 0.52–0.77 at $N \leq 260$, where the workhorse sits near zero). The conclusion is not an artifact of the estimator: CatBoost tracks slightly above the forest below full data (0.7667/0.8624/0.8915 seed means) and XGBoost slightly below it, so every classical variant we tested dominates the workhorse until labels are plentiful. Only at 6,589 days does the workhorse overtake them, and even there the identification gap to the strongest classical model is not statistically significant, as Section II-A established; the full-data edge that survives testing is binary specificity. In other words, a utility with few labeled days should train the forest, not the LLM. The case for fine-tuning rests on the full-data regime and on the capabilities the rest of this section establishes. However, the more mechanistically interesting finding is in the binary columns. At $N \leq 260$ the model is a near-constant *manipulated* predictor (specificity 0.008–0.092) even though its AUC has already reached 0.887–0.918. The ranking is learned almost immediately; the decision boundary is what takes thousands of examples. This mirrors the zero-shot observation of Section II-A from the opposite direction. It also suggests a practical shortcut we did not pursue: a utility with few labels could combine the early-learned ranking with a calibrated threshold instead of learning the full boundary. In other words, most of the labeled data buys calibration, not perception. The distinction is operational, not statistical — an uncalibrated detector at this specificity would dispatch an inspection to almost every consumer.

Identification follows the same order, and earlier than expected. At 65 training days the model already names the correct attack on 35.5% of manipulated days — more than four times the 8.3% chance rate — while its detector is still a near-constant; at 260 days the rate reaches 46.0%. The model learns to name attacks before it learns to doubt them, presumably because the benign class is one of thirteen and

TABLE VI

OUT-OF-CLASS BINARY DETECTION PER HELD-OUT FAMILY ROTATION, EVALUATED ON THE HELD-OUT ATTACKS PLUS ALL 130 BENIGN DAYS, FOR THE WORKHORSE AND THE MATCHED RANDOM-FOREST REFIT. HELD-OUT ATTACKS: R-Z = 0, 2, 13; R-M = 1, 3, 4, 5; R-T = 6, 7; R-S = 9, 10, 12.

Rot.	Detector	n	Bal-acc	F1	AUC	Spec.	Recall
R-Z	Workhorse	520	0.9179	0.9562	0.9999	0.8846	0.9513
	RF (refit)	520	0.8603	0.9546	0.9918	0.7231	0.9974
R-M	Workhorse	635	0.8874	0.9137	0.9435	0.9154	0.8594
	RF (refit)	635	0.8529	0.8761	0.9192	0.9077	0.7980
R-T	Workhorse	365	0.9325	0.9630	0.9910	0.8692	0.9957
	RF (refit)	365	0.7973	0.8584	0.9203	0.7308	0.8638
R-S	Workhorse	520	0.8462	0.8497	0.9171	0.9385	0.7538
	RF (refit)	520	0.8346	0.9312	0.9379	0.7154	0.9538

calibrating against it requires benign contrast the small subsets barely contain. Notably, format is learned before either: every prediction at every training size parses against the output contract, with zero schema violations even at 65 days. This ordering has a deployment reading. The capability the cascade of Section II-G buys from the LLM — adjudicating and naming days a cheap screen has already flagged — is precisely the one that emerges first, so even a label-poor utility could pair a classical screen with a lightly trained adjudicator. In other words, what labeled data buys is, in order, format, then taxonomy, then doubt. In summary, we can conclude that the low-label regime belongs to the classical model, and that the workhorse’s parity is purchased with the full 6,589 labeled days and \$13 of compute. However, every day evaluated so far exhibits an attack each detector saw in training, an assumption no deployment can rely on, and the next experiment removes it — for both of them.

C. Generalization to Unseen Attack Families (RQ3)

Having established that the recipe is affordable, we now remove its most optimistic assumption: that the attack catalog at training time is complete. Real manipulation strategies evolve. We therefore retrain the workhorse four times, each time holding out one entire attack family from Table I — zero-suppression (R-Z), multiplicative (R-M), threshold (R-T), and shape substitution (R-S) — and evaluate on the held-out attacks plus all 130 benign days. The attack descriptions remain in the prompt catalog, so the rotations also test whether a described-but-never-demonstrated attack can be recognized. Table VI reports the per-rotation aggregates, Table VII the recall on each never-seen attack, and Figure 4 the same per-attack view graphically.

Detection transfers remarkably well. Ten of the twelve never-seen attack types are detected at a recall of 0.85 or higher, five of them at 1.000. Specificity on the benign days is preserved in every rotation (0.869–0.938), so the transfer is not bought with false alarms. The two failures are attack4, a low-severity reduction confined to a sub-window ($\bar{\rho} = 0.084$, recall 0.437), and attack10, the energy-neutral time reversal ($\bar{\rho} = 0.043$, recall 0.269). These two attacks share no family — one is multiplicative, one is shape substitution — but they

TABLE VII

DETECTION RECALL ON EACH NEVER-SEEN ATTACK TYPE, WITH THE MEAN REALIZED ENERGY-REMOVAL PROXY $\bar{\rho}$. THE TWO FAILURES, MARKED *, ARE THE TWO STATISTICALLY QUIET ATTACKS.

Rotation	Attack	n	Recall	$\bar{\rho}$
R-Z	attack0	130	1.000	1.000
	attack2	130	1.000	0.136
	attack13	130	0.854	0.501
R-M	attack1	126	0.968	0.521
	attack3	130	1.000	0.554
	attack4*	119	0.437	0.084
	attack5	130	1.000	0.500
R-T	attack6	130	0.992	0.372
	attack7	105	1.000	0.649
R-S	attack9	130	1.000	0.043
	attack10*	130	0.269	0.043
	attack12	130	0.992	0.618

*Statistically quiet: $\bar{\rho} < 0.09$ (attack10 is energy-neutral by construction).

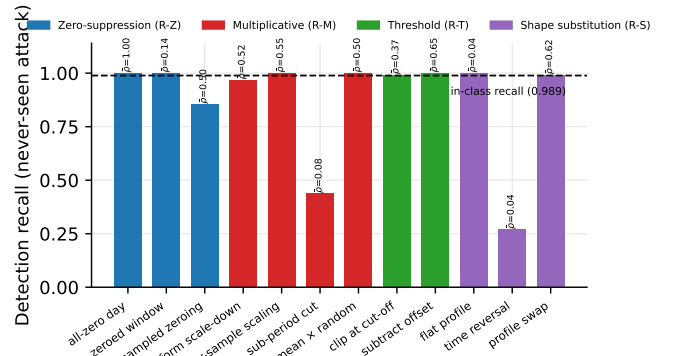


Fig. 4. Detection recall per never-seen attack, grouped by held-out rotation and annotated with the mean energy-removal proxy $\bar{\rho}$. The horizontal line is the in-class recall of the workhorse (0.989).

share a statistical property: both barely move the day’s energy. This suggests that what generalizes is not knowledge of attack mechanisms but sensitivity to removed energy. The detection floor is set by the magnitude of the theft, not by its type.

Identification is the opposite story. Multiclass accuracy on held-out attacks is 0.000 in every rotation except R-Z, where attack2 is partially recovered from its prompt description (precision 1.000, recall 0.292). We note that rotation-level macro-F1 values (0.054–0.102) are an artifact of nine to eleven zero-support classes and should not be read as partial success. Naming an attack, unlike detecting it, evidently requires demonstrations; the prompt description alone recovers a usable signal for only the most distinctive mechanism. For a utility this asymmetry is acceptable. The operational action (inspect) follows from detection, while identification informs the follow-up investigation.

The same stress test applied to the classical baseline re-frames the finding. Refit per rotation on the same reduced catalogs, the random forest also transfers detection (recall 0.798–0.997), but it pays in specificity: 0.715–0.731 on three of the four rotations, against the workhorse’s 0.869–0.938. The per-attack pattern is the more instructive part. On attack4 the forest fails harder than the LLM (recall 0.227 against 0.437), so the quiet-theft floor is shared where the energy signal is all

TABLE VIII
DETECTION RECALL BY THEFT-SEVERITY BUCKET (ρ EDGES AT 10/30/60% OF ENERGY REMOVED) ON THE EVALUATION SLICE. THE SEVERITY GRADIENT IS CONFINED ALMOST ENTIRELY TO THE LIGHTEST BUCKET.

Model	<10% (n=138)	10–30% (n=195)	30–60% (n=529)	>60% (n=398)	Energy-neutral (n=260)
Qwen3-8B	0.957	0.990	0.998	1.000	0.969
Qwen3-14B	0.964	1.000	1.000	1.000	0.977
Phi-4-reasoning	0.971	1.000	0.998	1.000	0.977
Ministral-3-14B	0.971	1.000	0.998	1.000	0.965
Llama-3.1-8B	0.964	1.000	1.000	1.000	0.973
Gemma-4-E4B	0.884	0.995	1.000	1.000	0.969
Gemma-4-31B	0.768	0.964	0.996	1.000	0.954
Qwen3-8B (magnitude)	0.949	1.000	1.000	1.000	0.981

there is. On attack10, however, the forest reaches 0.862 where the LLM managed 0.269 — its explicit shape-correlation feature recovers the time reversal — while on attack6 the roles invert (0.754 against 0.992). This suggests that the detection floor has two components: a shared one, set by how little energy the attack moves, and a detector-specific one, set by what each model’s representation happens to encode. The pattern is estimator-robust: CatBoost and XGBoost refits reproduce all three signatures (attack4 at 0.286 and 0.235, attack10 at 0.946 and 0.846, attack6 at 0.715 for both; full grid in Appendix D). In other words, the classical family and the LLM fail in different places, which is precisely the property a cascade can exploit.

One further regularity in the rotation table deserves a sentence. The workhorse’s specificity barely moves as the training catalog changes — 0.869–0.938 across the four rotations, against 0.892 with the full catalog — so its false-alarm rate is evidently a property of its model of the benign consumer, not of which attacks it has seen. This suggests that catalog updates, including retraining on newly discovered attack types, can be deployed without re-budgeting the inspection workload, since the false-positive side of the detector is anchored in what does not change. In summary, we can conclude that detection survives novel attack families wherever meaningful energy is removed, for both detectors. The LLM’s two failures point at a single governing variable, which the next experiment measures directly.

D. Severity-Stratified Failure Analysis (RQ4)

The out-of-class failures suggested that removed energy, not attack family, governs detectability. This experiment formalizes that hypothesis by stratifying every model’s recall over the severity proxy ρ of Section I-E; it requires no new training. Table VIII reports recall per severity bucket for the seven screened models and the magnitude variant of Section II-F, with the energy-neutral pair (attacks 9 and 10, which leave the day’s total energy unchanged and alter only the profile) reported separately. Figure 5 plots the curves. The full table including multiclass accuracy and the low-data runs is in Appendix C.

The hypothesis holds with unusual clarity. Every fine-tuned model reaches a recall of 1.000 in the >60% bucket, and the entire quality gradient between models is compressed into the <10% bucket, where the best models hold 0.971

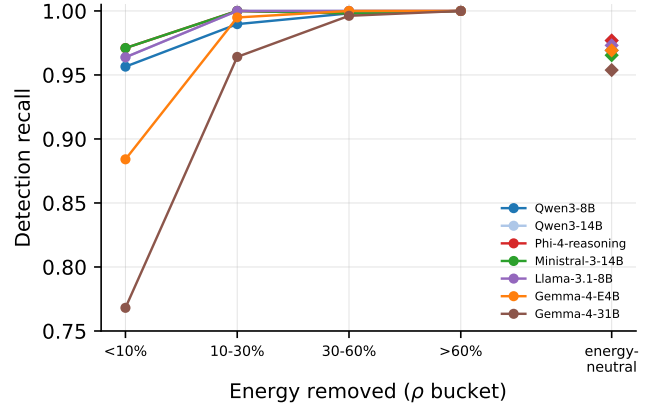


Fig. 5. Detection recall against theft severity for the screened models, with the energy-neutral pair (total energy unchanged, profile reshaped) as detached markers. Every model saturates above 60% energy removal; the differences live below 10%.

and Gemma-4-31B drops to 0.768. The energy-neutral pair, invisible to any energy-based criterion, is nevertheless detected at 0.954–0.981. This indicates that the models do exploit profile shape and not the energy total alone. For the utility, the reading is direct. Theft large enough to matter financially is detected essentially always, and the residual risk is confined to manipulations that remove less than a tenth of a day’s energy — precisely the manipulations whose per-day revenue impact is smallest. However, this finding sharpens rather than settles the objection raised at the end of Section II-A. The random forest also detects consequential theft, so if detection saturates for every competent model, the case for an LLM must rest on what it produces beyond the flag. In summary, we can conclude that severity, not attack family, is the axis along which detectors fail, and we turn to the capability that classical baselines do not offer at all.

E. Rationale Distillation (RQ5)

A flagged day triggers a human decision, and the investigator’s first question is not *whether* but *what happened*. This experiment asks whether the workhorse can be taught to answer it, and what that costs in classification quality. We fine-tune four variants on teacher-generated rationales under a catalog-free prompt, so the attack knowledge must live in the weights rather than the prompt. R0 is a label-only control. R1 trains on rationale plus label over the full 6,589 days. R2a and R2b are a matched 709-day pair contrasting a label-aware teacher, which knows the true attack when writing the rationale, against a label-blind one. Table IX reports classification and explanation quality; Table X the pre-registered paired contrasts.

Three findings emerge. First, explanations are not free, but they are cheap. Training on rationales costs -0.0159 macro-F1 against the label-only control, a small but statistically significant identification penalty (CI excluding zero; McNemar $p = 0.0067$). Binary detection, however, is statistically unchanged (Δ balanced accuracy $+0.0029$, CI containing zero), so the utility pays nothing in detection to gain an

TABLE IX

RATIONALE DISTILLATION ON THE WORKHORSE UNDER THE CATALOG-FREE PROMPT: CLASSIFICATION, EXPLANATION OVERLAP (ROUGE-L), AND PARAMETER-EXTRACTION FIDELITY. R0’S MACRO-F1 CARRIES ‡: IT STEMS FROM A DIFFERENT RECIPE THAN THE SCREENING RUNS AND IS NOT COMPARABLE TO TABLE III.

Run	Train	Macro-F1	Bin. bal-acc	Spec.	ROUGE-L	Param. correct
R0 (label only)	6,589	0.9444 ‡	0.9384	0.885	—	—
R1 (rationale+label)	6,589	0.9285	0.9412	0.892	0.454	0.853
R2a (label-aware, small)	709	0.0715	0.5705	0.169	0.289	0.277
R2b (label-blind, small)	709	0.0821	0.6074	0.262	0.305	0.400

Parameter extraction is objectively checkable for attacks 1, 2, and 4 only ($n=375$, $\approx 39\%$ of manipulated days).

TABLE X

PAIRED CONTRASTS FOR THE RATIONALE-DISTILLATION QUESTIONS: CONSUMER-CLUSTER BOOTSTRAP CIs AND McNEMAR TESTS ON IDENTICAL PREDICTIONS.

Comparison	Δ macro-F1	95% CI	Δ bin. bal-acc	McNemar p (disc.)
R1 — R0	−0.0159	[−0.0284, −0.0037]	+0.0029	0.0067 (93)
R2b — R2a	+0.0106	[−0.0001, +0.0216]	+0.0369	0.851 (113)

explanation for every flag. Second, the explanations are substantive rather than decorative. R1 reaches a ROUGE-L of 0.454 against the teacher and, on the stricter test, states the numerically correct attack parameter on 85.3% of checkable days. Third, the label-blind teacher supervises explanations as well as the label-aware one. R2b is statistically tied with R2a on classification and ahead on parameter fidelity (0.400 vs. 0.277). Notably, the label-aware teacher was handed the exact generation parameters when writing its rationales, yet its student recovers the correct parameter *less* often than the student of the teacher that never saw them. This suggests that label-aware rationales transmit the teacher’s conclusion rather than the evidence-reading that should support it: a teacher forced to infer the attack from the readings must demonstrate the derivation, and the derivation, not the verdict, is what a student can reuse on unseen days. In other words, rationale supervision transfers a skill, not an answer — and only the label-blind teacher exercises the skill. Beyond explaining the R2 ordering, this removes a deployment dependency, since rationale supervision evidently does not require annotated attacks at teacher time. We note two scope limits honestly. The planned LLM-judge assessment of rationale quality was not run; a frozen 1,440-row judging package with a pinned rubric is prepared for external scoring. R0’s headline macro-F1 of 0.9444, the highest in this paper, comes from a different recipe than the screening runs and therefore does not enter the model comparison of Section II-A. An explanation names the mechanism; an inspector planning recovery also needs its size, which is the final capability we test. In summary, we can conclude that mechanism explanations cost 1.6 macro-F1 points and no detection, and we next extend the same output schema from naming the theft to quantifying it.

F. Quantifying Theft Magnitude (RQ5)

Extending explanation to quantification, we append a magnitude block to the output schema: the estimated fraction of energy removed and the manipulated time window. The workhorse is fine-tuned to emit it alongside the classification, with ground truth recomputed from the readings via the

TABLE XI

EXPLICIT MAGNITUDE QUANTIFICATION BY THE WORKHORSE: HEADLINE ERROR, CALIBRATION, AND LOCALIZATION (TOP), AGAINST PER-DAY CONTROLS AND THE READING-LEVEL CEILING (BOTTOM). MAE IS ON THE ENERGY-REDUCTION FRACTION $\rho \in [0, 1]$, $n = 1,260$ MAGNITUDE-BEARING DAYS.

Quantity	Value		
MAE (95% CI)	0.0321 [0.0286, 0.0369]		
Pearson / Spearman	0.985 / 0.976		
Severity-bucket accuracy	0.882		
Window coverage / mean IoU	0.956 / 0.874		
Schema-violation rate	0.000		
Control	MAE	RMSE	95% CI
Workhorse (LLM)	0.0321	0.0508	[0.0286, 0.0369]
Class-prior (predicted class)	0.0897	0.1378	[0.0829, 0.0974]
Class-prior (gold class)	0.0903	0.1385	[0.0834, 0.0981]
Ridge on readings	0.1313	0.1602	[0.1247, 0.1391]
GBM on readings (ceiling)	0.0012	0.0020	[0.0011, 0.0013]

severity proxy. Two questions matter. The first is whether the estimates are genuine per-day measurements rather than class-conditional averages; the second is whether carrying the extra target degrades classification. Table XI answers the first; the pre-registered interference check answers the second.

The estimates are genuine. The model attains an MAE of 0.0321 on the energy-reduction fraction with a Spearman correlation of 0.976. It beats the class-prior control — the strategy of looking up the average severity of the predicted attack class — by 0.0576 MAE with a confidence interval excluding zero. The same holds against the prior computed from the gold class, so the advantage is not explained by classification accuracy. Localization is similarly strong (window IoU 0.874, boundary error under one 15-minute interval). The model correctly declines to claim magnitude on the energy-neutral attacks (mean claimed ρ of 0.012 against a true 0.043), and the schema is never violated. However, a gradient-boosted regressor reading the same raw values attains an MAE of 0.0012, roughly 27 times lower. For magnitude in isolation, the readings themselves remain the accuracy ceiling and a dedicated regressor the better tool. The pre-registered interference check completes the picture: adding the magnitude target does not degrade classification relative to the plain workhorse (Δ macro-F1 +0.0057, CI [−0.0043, +0.0157]; McNemar $p = 1.000$ binary, 0.295 multiclass). Set against Section II-E, this yields a pattern worth stating: the numerically harder auxiliary task (per-day magnitude regression) costs classification nothing, while the linguistically heavier one (free-text rationales) cost a small but significant 1.6 macro-F1 points. This suggests that the price of an auxiliary target tracks the token mass it adds to the training signal rather than the difficulty of the task — the magnitude block is a handful of tokens, a rationale is a paragraph — and the training losses point the same way (0.014 label-only against 0.42 with rationales). We state this as a reading of two data points, not a law. In other words, the model’s value is not any single number but the integrated output — one JSON object that detects the theft, names its mechanism, estimates its size, and locates its window, at no measured cost to the detection it rides on. With the capability set now complete, what remains is to price it. In

TABLE XII

THE DEPLOYMENT FRONTIER AT ASSUMED PRICES: IDENTIFICATION QUALITY AGAINST INFERENCE COST PER 1,000 QUERY-DAYS. PARETO-OPTIMAL POINTS ARE BOLDDED; THE CASCADE USES THE RANDOM FOREST TO SCREEN AND THE WORKHORSE TO ADJUDICATE FLAGGED DAYS.

Operating point	Kind	Macro-F1	\$/1k days	s/day
Random forest (refit)	classical, CPU	0.9122	0.01	—
Qwen3-8B (workhorse)	local LLM	0.9229	7.48	10.8
Ministral-3-14B	local LLM	0.9298	9.07	13.1
Phi-4-reasoning (flagship)	local LLM	0.9331	9.07	13.1
Gemma-4-31B	local LLM	0.8517	33.89	48.8
Gemini 3.1 Pro	API, zero-shot	0.929	32.50	—
Cascade (RF → workhorse)	hybrid	0.9114	6.92	—

Flagship throughput estimated from the same-size Ministral-3-14B; all costs assumption-driven.

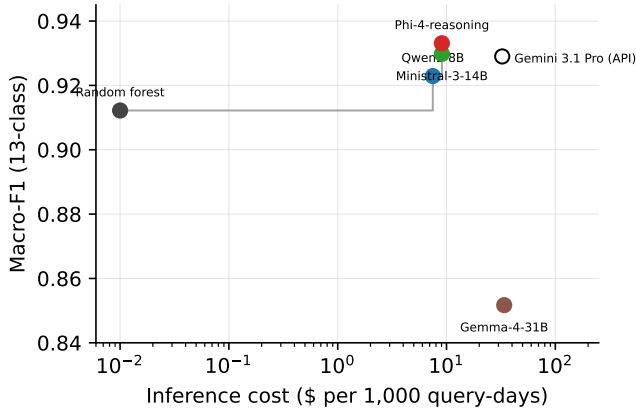


Fig. 6. The cost–performance frontier: macro-F1 against inference cost per 1,000 query-days (log scale). The step line traces the Pareto front; the open marker is the zero-shot API point. The cascade operating point is reported in the frontier table.

summary, we can conclude that magnitude estimation is real but ceiling-limited, and the deployment analysis that follows weighs the whole package against its alternatives.

G. Cost–Performance Deployment Frontier (RQ6)

The preceding sections established what the fine-tuned detector achieves, tolerates, and produces. This final experiment assembles those results, at assumed prices, into the deployment decision a utility actually faces. Table XII and Figure 6 place every operating point on a cost–accuracy plane: the refit random forest on CPU, the fine-tuned models on an amortized local H100 (\$2.50/h), and the zero-shot API reference (\$5 per million tokens, ~6,500 tokens per query-day).

The Pareto front runs random forest → workhorse → flagship, each step buying roughly one macro-F1 point for the next order of cost. Both the API reference and the 31B model sit strictly inside the front. The API costs 4.3 times the workhorse for 0.007 more macro-F1, and the 31B model costs more than the API while identifying worst of all. Against the API, the local workhorse breaks even after 523 query-days, its one-off \$13.08 fine-tune being repaid at \$0.025 per query-day thereafter. For any utility screening even a modest fleet daily, this is hours, not months. The frontier prices inference only, so the label bill of Section II-B must be read alongside it: every fine-tuned point on the frontier was bought with the same

6,589 labeled days, and a utility willing to accept 81% of the identification quality can enter the frontier at 1,040 labels and \$2.07 of training. The random forest consumes the identical labeled set, so at full data the frontier comparison is label-neutral. Below full data it is not: Section II-B showed the forest degrading far more gracefully (0.745 macro-F1 at 65 days against the workhorse’s 0.259), so the low-label regime belongs to the classical screen outright. The complementary failure profiles of Section II-C are one more reason the cascade pairs the two detectors rather than choosing between them. The tension carried since Section II-A resolves here as a complement rather than a competition. In the cascade, the near-free random forest screens every day and the workhorse adjudicates only flagged ones, retaining 0.9114 macro-F1 at \$6.92 per thousand days. More to the point, the cascade attaches a mechanism explanation, a magnitude estimate, and a manipulation window to precisely the days a human will review. The LLM rides on infrastructure sized by the screening rate rather than the fleet, so this operating point scales with the inspection workload the utility already budgets for. However, the cascade’s flag rate on this theft-heavy evaluation slice (0.925) is far above what a deployed screening population would produce, so the quoted blended cost is an upper bound on the LLM share. In summary, we can conclude that the deployment argument for fine-tuned open models is not that they beat classical screening on accuracy, which they do only narrowly. It is that at \$7.48 per thousand days they convert each flag from a score into an actionable, explained, quantified finding, with the classical screen retained as the free first stage.

H. Threats to Validity and Limitations

While these experiments support the conclusions above, certain limitations should be acknowledged. Firstly, the frozen conference ML bands (CatBoost, XGBoost, LOF) were obtained on a different, 1,261-row benchmark. We treat them strictly as context bands and never test against them; the Gemini frontier, by contrast, is reported on the same split and carries no such caveat. Secondly, all attacks are synthetic manipulations of one real consumption dataset. The taxonomy is standard in the non-technical-loss literature, but real theft need not respect it, and our own out-of-class results show detection degrading precisely where an attack departs from the energy-removal pattern the taxonomy embodies. Thirdly, the classical baselines run at seeded library-default hyperparameters. All three tree-family estimators are fully matched across the full-data, data-size, and rotation grids, but the conference’s tuned CatBoost configuration is not recorded in our artifacts, so the classical numbers here are floors rather than tuned ceilings; the logistic regression and LOF remain balanced-accuracy only. Fourthly, the rationale evaluation is bounded by its instruments: ROUGE-L rewards overlap rather than truth, parameter fidelity covers 39% of manipulated days, and the planned LLM-judge assessment was not run. Fifthly, every deployment cost rests on assumed prices (amortized \$2.50/h GPU, blended API tokens), and a per-run hardware manifest exists only for the legacy pilot. The frontier’s ordering is robust to price changes, but its absolute dollar figures are estimates.

TABLE XIII
LoRA-RANK SWEEP AROUND THE SCREENING DEFAULT ON THE
WORKHORSE. PAIRED STATISTICS ARE AGAINST $r = 16$.

Rank	Macro-F1	Bin. bal-acc	AUC	Spec.	Δ vs. $r=16$	McNemar p
$r = 8$	0.9140	0.9454	0.9943	0.900	-0.0089	0.101
$r = 16$ (default)	0.9229	0.9406	0.9939	0.892	—	—
$r = 32$	0.9073	0.9444	0.9930	0.900	-0.0156	0.0158

Finally, the detection floor identified in Sections II-C and II-D — statistically quiet manipulation — is a structural limitation of consumption-based detection, not of one model. Adversaries aware of it could deliberately operate below the 10% band, at proportionally reduced benefit. Future work will re-score the legacy 32B pilot on this benchmark, obtain external judge scores from the frozen package, and fill the classical grid cells. None of these threatens the funnel’s direction, but each would tighten a bound this paper leaves loose.

APPENDIX A RECIPE VALIDATION: LoRA RANK

The screening of Section II-A applied one shared recipe to all seven models, which is only fair if that recipe was not capacity-limiting. We validate this on the workhorse by sweeping the LoRA rank one step below and above the default. Table XIII shows that halving the rank is statistically indistinguishable from the default (Δ macro-F1 -0.0089 , 95% CI $[-0.0197, +0.0015]$, McNemar $p = 0.101$). Doubling it is significantly *worse* ($\Delta = -0.0156$, CI $[-0.0277, -0.0037]$, $p = 0.0158$). The results suggest that the shared recipe sat at or slightly above the capacity the task requires, so no screened model was starved by it.

APPENDIX B PER-CLASS RESULTS

Table XIV reports per-class F1 for all seven fine-tuned models with the best zero-shot model as the contrast row; Table XV the per-class F1 of every zero-shot base; Figure 7 the flagship’s confusion matrix. Across every fine-tuned model, attack1 (uniform multiplicative reduction) and attack12 (profile substitution) are the two limiting classes, while attack0 and attack13 are essentially solved.

APPENDIX C SEVERITY STRATIFICATION, ALL RUNS

Table XVI extends Table VIII to every prediction set in the paper, reporting detection recall and multiclass accuracy per severity bucket. Two patterns are worth noting beyond the main text. First, the low-data runs ($N \leq 260$) show high recall in every bucket only because they flag nearly everything; compare their specificity in Table V. Their multiclass accuracy collapses instead. Second, the small rationale runs (R2a/R2b) preserve detection recall while identification collapses. This is consistent with 709 days being enough to keep the boundary but not the taxonomy.

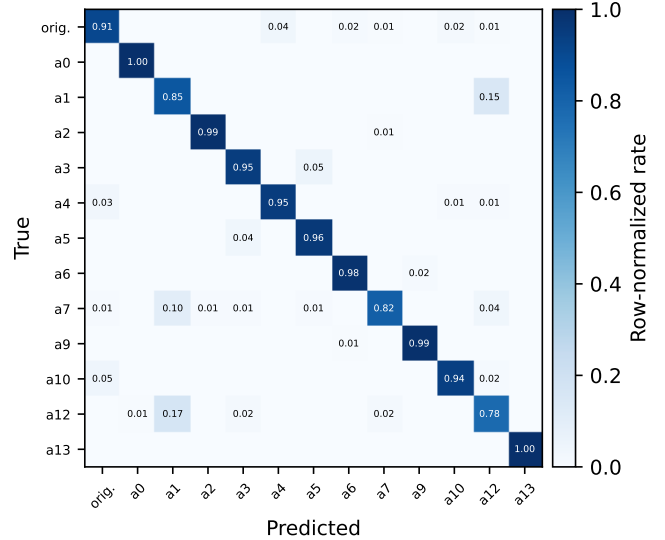


Fig. 7. Row-normalized confusion matrix of the flagship (Phi-4-reasoning) on the evaluation slice. The dominant confusions run between attack1 and the benign class, and between attack12 and the benign class.

APPENDIX D CLASSICAL GRID, ALL ESTIMATORS

Tables XVII and XVIII report the full classical grid — random forest, CatBoost, and XGBoost at seeded library defaults — on the matched data-curve subsets and rotation refits. Three observations carry to the main text. First, the ordering below full data is CatBoost > random forest > XGBoost, and every estimator dominates the workhorse until full data. Second, at full data the ordering inverts and the random forest is the classical ceiling (0.9122), which is why the paired tests of Section II-A target it. Thirdly, the rotation signatures — attack4 missed, attack10 caught, attack6 weakened — appear in all three estimators, so the two-component floor of Section II-C describes the feature family, not one model.

APPENDIX E EXPLANATION QUALITY ACROSS ALL MODELS

Table XIX reports ROUGE-L, parameter-mention rate, and parameter-correctness for every scored run. Notably, the screening models — trained with the attack catalog in the prompt — produce rationales of the same ROUGE-L band (0.43–0.46) as the dedicated R1 distillation run. This suggests that with catalog-prompt SFT, explanation ability comes essentially free; the catalog-free R1 run matters because it internalizes that ability into the weights. In every scored run, rationales for label-blind teacher rows score higher than for label-aware ones.

APPENDIX F TRAINING COST

Table XX records the training cost of every run, from the per-run metadata. All runs execute on a single H100 80 GB; the amortized figures of Section II-G derive from these wall-clock times at \$2.50/h. The 4-bit 31B model is the outlier at

TABLE XIV
PER-CLASS F1 ON THE EVALUATION SLICE FOR THE SEVEN FINE-TUNED MODELS, WITH THE BEST ZERO-SHOT MODEL (GEMMA-4-31B) AS THE BOTTOM ROW. BEST VALUE PER CLASS COLUMN IS BOLDDED WITHIN THE FINE-TUNED ROWS.

Model	orig.	a0	a1	a2	a3	a4	a5	a6	a7	a9	a10	a12	a13	Macro
Phi-4-reasoning	.911	.996	.805	.992	.939	.954	.951	.981	.878	.989	.953	.783	1.00	.933
Ministral-3-14B	.909	.996	.795	.996	.945	.953	.955	.969	.880	.977	.938	.773	1.00	.930
Qwen3-8B	.882	.996	.783	1.00	.929	.917	.943	.981	.826	.988	.941	.811	1.00	.923
Llama-3.1-8B	.873	.996	.763	.992	.941	.899	.943	.977	.852	.985	.953	.762	1.00	.918
Qwen3-14B	.915	.996	.765	.988	.929	.928	.940	.969	.836	.973	.946	.739	1.00	.917
Gemma-4-E4B	.865	.996	.622	.984	.927	.871	.924	.965	.836	.981	.949	.626	.996	.888
Gemma-4-31B	.804	.996	.647	.996	.808	.765	.835	.973	.791	.985	.906	.566	1.00	.852
Gemma-4-31B (zero-shot)	.762	.996	.613	.689	.694	.539	.248	.670	.000	.815	.750	.000	.929	.593

TABLE XV
PER-CLASS F1 OF EVERY ZERO-SHOT BASE MODEL ON THE EVALUATION SLICE.

Base model	orig.	a0	a1	a2	a3	a4	a5	a6	a7	a9	a10	a12	a13	Macro
Qwen3-8B	.146	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.011
Qwen3-14B	.147	.206	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.027
Phi-4-reasoning	.146	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.011
Ministral-3-14B	.217	.869	.390	.382	.030	.017	.000	.000	.000	.766	.015	.000	.478	.243
Llama-3.1-8B	.040	.743	.152	.405	.000	.180	.292	.124	.000	.733	.015	.000	.043	.210
Gemma-4-E4B	.027	.594	.298	.616	.000	.377	.028	.015	.000	.780	.015	.000	.868	.278
Gemma-4-31B	.762	.996	.613	.689	.694	.539	.248	.670	.000	.815	.750	.000	.929	.593

22.4 hours — quantized training trades memory for time — and the small rationale runs complete in half an hour.

APPENDIX G PER-ATTACK MAGNITUDE ERROR

Table XXI decomposes the magnitude MAE of Section II-F per attack. The total bypass (attack0), whose true ρ is identically 1, is estimated exactly. The hardest attacks for magnitude are those whose reduction is confined to sub-windows or applied irregularly (attack4, attack13). On the energy-neutral pair the model claims a mean ρ of 0.012 against a true 0.043; it correctly declines to assert magnitude where there is none to measure. On the 130 benign days it never emits a magnitude block for a day it classifies as original, and its 18 false positives carry a mean claimed ρ of 0.067.

APPENDIX H REMARK ON THE LEGACY 32B PILOT

An earlier pilot fine-tuned Qwen3-32B with 4-bit QLoRA under the project’s original data regime. It was evaluated on a 130-day class-stratified subset of a different holdout, with decision thresholds calibrated on a validation split. Its results (binary balanced accuracy up to 0.977 at a specificity of 1.0 under verbalizer thresholding; multiclass macro-F1 0.833) are not comparable to any table in this paper: different evaluation set, different size, and threshold tuning that the journal-round runs deliberately forgo. We cite them only to acknowledge the pilot’s existence. Re-scoring this checkpoint on the present benchmark is part of the future work listed in Section II-H.

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TABLE XVI
DETECTION RECALL / MULTICLASS ACCURACY PER SEVERITY BUCKET FOR EVERY PREDICTION SET (EVALUATION SLICE; BUCKET $n = 138/195/529/398$, ENERGY-NEUTRAL $n = 260$).

Run	<10%	10–30%	30–60%	>60%	Energy-neutral
Qwen3-8B (workhorse)	0.957 / 0.942	0.990 / 0.923	0.998 / 0.904	1.000 / 0.940	0.969 / 0.950
Qwen3-14B	0.964 / 0.913	1.000 / 0.928	1.000 / 0.898	1.000 / 0.912	0.977 / 0.965
Phi-4-reasoning (flagship)	0.971 / 0.957	1.000 / 0.944	0.998 / 0.923	1.000 / 0.925	0.977 / 0.965
Ministral-3-14B	0.971 / 0.957	1.000 / 0.954	0.998 / 0.906	1.000 / 0.930	0.965 / 0.954
Llama-3.1-8B	0.964 / 0.949	1.000 / 0.949	1.000 / 0.906	1.000 / 0.910	0.973 / 0.962
Gemma-4-E4B	0.884 / 0.848	0.995 / 0.923	1.000 / 0.856	1.000 / 0.872	0.969 / 0.962
Gemma-4-31B	0.768 / 0.761	0.964 / 0.856	0.996 / 0.828	1.000 / 0.837	0.954 / 0.938
Qwen3-8B (magnitude)	0.949 / 0.920	1.000 / 0.949	1.000 / 0.915	1.000 / 0.947	0.981 / 0.962
$r = 8$	0.949 / 0.928	1.000 / 0.918	0.998 / 0.883	1.000 / 0.930	0.977 / 0.954
$r = 32$	0.928 / 0.906	0.990 / 0.923	0.996 / 0.881	1.000 / 0.910	0.989 / 0.958
R0 (label only)	0.957 / 0.949	1.000 / 0.954	1.000 / 0.936	1.000 / 0.957	0.977 / 0.969
R1 (rationale+label)	0.942 / 0.935	1.000 / 0.949	0.996 / 0.913	1.000 / 0.935	0.981 / 0.954
R2a (709, label-aware)	0.986 / 0.007	1.000 / 0.113	0.947 / 0.144	0.993 / 0.317	0.962 / 0.000
R2b (709, label-blind)	0.964 / 0.087	0.990 / 0.164	0.938 / 0.136	0.977 / 0.251	0.915 / 0.000
$N=1,040$	0.928 / 0.804	0.990 / 0.856	1.000 / 0.667	0.995 / 0.754	0.954 / 0.896
$N=260$ (s42)	0.993 / 0.399	0.985 / 0.523	0.945 / 0.250	0.995 / 0.412	0.996 / 0.946
$N=260$ (s43)	0.978 / 0.370	0.990 / 0.518	0.896 / 0.268	0.980 / 0.410	0.973 / 0.858
$N=260$ (s44)	0.993 / 0.478	0.990 / 0.600	0.928 / 0.174	0.982 / 0.417	0.969 / 0.846
$N=65$ (s42)	0.978 / 0.203	0.969 / 0.380	0.890 / 0.113	0.970 / 0.427	0.989 / 0.800
$N=65$ (s43)	0.935 / 0.174	0.949 / 0.339	0.964 / 0.187	0.982 / 0.455	0.965 / 0.658
$N=65$ (s44)	0.935 / 0.138	0.923 / 0.313	0.915 / 0.176	0.965 / 0.455	0.962 / 0.712

TABLE XVIII
CLASSICAL ROTATION REFITS: OUT-OF-CLASS BINARY DETECTION PER HELD-OUT FAMILY (EVALUATED ON HELD-OUT ATTACKS PLUS ALL 130 BENIGN DAYS).

Rot.	Estimator	Bal-acc	Spec.	Recall
R-Z	Random forest	0.8603	0.7231	0.9974
	CatBoost	0.8218	0.6462	0.9974
	XGBoost	0.8385	0.6769	1.0000
R-M	Random forest	0.8529	0.9077	0.7980
	CatBoost	0.8617	0.9154	0.8079
	XGBoost	0.8353	0.8923	0.7782
R-T	Random forest	0.7973	0.7308	0.8638
	CatBoost	0.7520	0.6615	0.8426
	XGBoost	0.7478	0.6615	0.8340
R-S	Random forest	0.8346	0.7154	0.9538
	CatBoost	0.8487	0.7154	0.9821
	XGBoost	0.8038	0.6615	0.9462

TABLE XVII
CLASSICAL DATA-CURVE CELLS: MACRO-F1 PER TRAINING-SET SIZE AND DATA SEED (MATCHED SUBSETS).

N	Seed	Random forest	CatBoost	XGBoost
65	42	0.7905	0.8126	0.7180
	43	0.7327	0.7587	0.6459
	44	0.7120	0.7288	0.7118
260	42	0.8473	0.8706	0.8417
	43	0.8324	0.8675	0.8245
	44	0.8167	0.8491	0.8477
1,040	42	0.8855	0.8915	0.8790
6,589 (full)	—	0.9122	0.9093	0.9076

TABLE XIX

EXPLANATION QUALITY FOR EVERY SCORED PREDICTION SET ($n = 1,520$ MANIPULATED DAYS; PARAMETER EXTRACTION ON THE CHECKABLE SUBSET, $n = 375$).

Run	ROUGE-L	Param. mentioned	Param. correct
R0 (label only)	0.000	0.000	0.000
R1 (rationale+label)	0.454	0.931	0.853
R2a (709, label-aware)	0.289	0.949	0.277
R2b (709, label-blind)	0.305	0.973	0.400
Qwen3-8B (workhorse)	0.455	0.925	0.821
Qwen3-14B	0.456	0.915	0.680
Phi-4-reasoning	0.457	0.952	0.845
Ministral-3-14B	0.462	0.920	0.619
Llama-3.1-8B	0.453	0.917	0.632
Gemma-4-E4B	0.444	0.829	0.712
Gemma-4-31B	0.431	0.824	0.629
Qwen3-8B (magnitude)	0.458	0.944	0.843

TABLE XXI

MAGNITUDE MAE PER ATTACK (MAGNITUDE-BEARING ATTACKS; n PER CLASS AS IN SECTION I-A).

Attack	MAE (ρ)
attack0	0.0000
attack3	0.0272
attack5	0.0279
attack1	0.0291
attack7	0.0300
attack12	0.0300
attack2	0.0419
attack6	0.0426
attack4	0.0454
attack13	0.0473

TABLE XX

TRAINING COST PER RUN: TRAINING ROWS, WALL-CLOCK, AND FINAL TRAINING LOSS.

Run	Train rows	Wall-clock (h)	Final loss
Qwen3-8B (workhorse)	6,589	5.23	0.407
Qwen3-14B	6,589	8.29	0.387
Phi-4-reasoning	6,589	5.52	0.424
Ministral-3-14B	6,589	7.52	0.354
Llama-3.1-8B	6,589	4.12	0.410
Gemma-4-E4B	6,589	7.43	0.338
Gemma-4-31B (4-bit)	6,589	22.36	0.373
Qwen3-8B (magnitude)	6,589	5.43	0.371
$r = 8 / r = 32$	6,589	5.24 / 5.23	0.406 / 0.408
R-Z / R-M / R-T / R-S	5,035–5,646	3.6–4.5	0.39–0.44
R0 / R1	6,589	4.62 / 4.73	0.014 / 0.424
R2a / R2b	709	0.51 / 0.51	0.670 / 0.724
$N \in \{65, 260, 1,040\}$	65–1,040	0.05–0.83	0.56–1.25

R0's low loss (0.014) reflects label-only targets being near-trivially learnable; the gap to R1 is the cost of the rationale tokens, not quality.